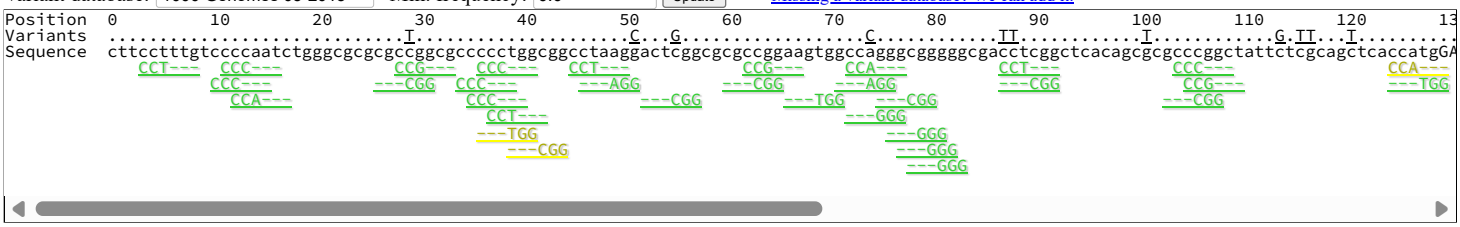
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Homo sapiens (hg19), chr7:5569177-5569415, reverse genomic strand
Your input sequence is 239 bp long. It contains 52 possible guide sequences.
Shown below are their PAM sites and the expected cleavage position located -3bp 5' of the PAM site.
Click on a match for the PAM NGG below to show its 20 bp-long guide sequence. (Need help? Look at the CRISPOR manual)
Colors green, yellow and red indicate high, medium and low specificity of the PAM's guide sequence in the genome.

Gene Models: Do not show Update
Variant database: 1000 Genomes 05-2013 Min. frequency: 0.0 Update Missing a variant database? We can add it.



Download for: SerialCloner (free) - ApE (free) - GenomeCompiler - Benchling - SnapGene - Geneious - Vector NTI - LaserGene - Genbank - FASTA

Predicted guide sequences for PAMs
Ranked by default from highest to lowest specificity score (Hsu et al., Nat Biot 2013). Click on a column title to rank by a score.
If you use this website, please cite our paper in NAR 2018. Too much information? Look at the CRISPOR manual.

Download as Excel tables: Guides / Guides, all scores / Off-targets / Saturating mutagenesis assistant

Position/ Strand	Guide Sequence + *PAM* + Restriction Enzymes + Variants Only G- Only GG- Only A-	MIT Specificity Score	CFD Spec. score	Predicted Efficiency Show all scores Doench '16	Mor.-Mateo	Doench-RuleSet3	Out-of-Frame	Lindel	Off-targets for 0-1-2-3-4 mismatches + next to PAM	Genome Browser links to matches sorted by CFD off-target score exons only chr7 only
46 / rev	TTCCGGCGCGCCGAGTCCTT *AGG* Inefficient Enzymes: *Bse21I, BshFI, SsiI, Fsp4HI, BstDEI, TauI* Cloning / PCR primers	97	98	47	57	-43	67	75	0 - 0 - 0 - 0 - 14 0 - 0 - 0 - 0 - 1 14 off-targets	4:intergenic:SYNJ2BP/SYNJ2 COX16-RP11-486O13.4 4:intergenic:AC013275.2-SCTR 4:intron:CR1L show all...
12 / rev	CGCCGGCGCGCGCCCAGATT *GGG* High GC content Enzymes: *BslFI, BslI* Cloning / PCR primers	96	99	22	58	-95	71	78	0 - 0 - 0 - 3 - 28 0 - 0 - 0 - 0 - 0 31 off-targets	4:exon:LMF1 4:intergenic:MIR4291-RP11-53B5.1 4:intron:DAZAP1 show all...
30 / fw	TCCCCAATCTGGGCGCGCGC *CGG* Enzymes: *MspI, LpnPI, HinP1I, BspFNI, AcyI, NlaIV, BstC8I, Cfr10I, BssHII, BfoI, NaeI, SspDI, BanI, SgrAI, MreI* Cloning / PCR primers	93	95	40	36	2	65	88	0 - 0 - 0 - 0 - 55 0 - 0 - 0 - 0 - 1 55 off-targets	4:intron:EPS15L1 4:exon:USP32 4:exon:SBF1 show all...
64 / fw	GGCCTAAGGACTCGGCGCGC *CGG* Enzymes: *MspI, LpnPI, HinP1I, BspFNI, BstC8I, BssHII, PalAI* Cloning / PCR primers	92	97	51	45	20	61	86	0 - 0 - 0 - 1 - 50 0 - 0 - 0 - 0 - 0 51 off-targets	4:intergenic:NKAIN1-SNRNP40 4:intergenic:SYNJ2BP/SYNJ2 COX16-RP11-486O13.4 4:intergenic:RP11-516C1.1/CASC4-RP11-516C1.1 show all...
38 / rev	CGCCGAGTCCTTAGGCCGCC *AGG* Enzymes: *LpnPI, SsiI, BseDI, Fsp4HI, BstNI, StyD4I, TauI, BslI* Cloning / PCR primers	91	92	31	40	-123	75	75	0 - 0 - 0 - 2 - 52 0 - 0 - 0 - 0 - 0 54 off-targets	3:intergenic:ISCA1P4-CHAC1 4:intron:AGAP3 4:intergenic:Y_RNA-CCDC85C show all...
163 / fw	GCCGCGCTCGTCGTCGACAA *CGG* Enzymes: *FblI, Hpy166II, NlaIV, HincII, SalI, BceAI, LmnI* Cloning / PCR primers	91	97	57	64	66	45	72	0 - 0 - 0 - 14 - 14 0 - 0 - 0 - 1 - 0 28 off-targets	3:exon:RP11-390F4.8 3:exon:ACTBP9 4:exon:ACTBP7 show all...
37 / rev	GCCGAGTCCTTAGGCCGCCA *GGG* Enzymes: *LpnPI, SsiI, HinP1I, BseDI, Fsp4HI, BstNI, StyD4I, TauI, BslI* Cloning / PCR primers	90	95	47	56	40	75	73	0 - 0 - 0 - 9 - 37 0 - 0 - 0 - 0 - 0 46 off-targets	3:intergenic:ISCA1P4-CHAC1 4:intergenic:ABCC6P1-RPS15A 3:intergenic:LINC00333-LINC00375 show all...
70 / fw	AGGACTCGGCGCGCCGGAAG *TGG* Enzymes: *LpnPI, BshFI, AcoI, BalI, BstNI, StyD4I* Cloning / PCR primers	90	96	47	61	21	58	71	0 - 0 - 0 - 6 - 57 0 - 0 - 0 - 0 - 1 63 off-targets	4:intergenic:RP11-104L21.3-CD247 4:intergenic:RP11-294L11.1-VIT 4:exon:PPP1R11 show all...

Position/ Strand	Guide Sequence + *PAM* + Restriction Enzymes + Variants Only G- Only GG- Only A-	MIT Specificity Score	CFD Spec. score	Predicted Efficiency Show all scores Doench '16	Mor.-Mateo	Doench-RuleSet3	Out-of-Frame	Lindel	Off-targets for 0-1-2-3-4 mismatches + next to PAM	Genome Browser links to matches sorted by CFD off-target score exons only chr7 only
75 / fw	TCGGCGCGCCGGAAGTGGCC *AGG* .G.....C.. ⚠Inefficient ⚠High GC content Enzymes: *LpnPI, BshFI, AcoI, BseDI, BalI, BstNI, StyD4I* Cloning / PCR primers	90	97	40	52	-44	56	73	0 - 0 - 1 - 4 - 28 0 - 0 - 1 - 2 - 1 33 off-targets	4:exon:CDK13 4:intron:ZSWIM8 4:exon:C9orf78 show all...
107 / fw	ACCTCGGCTCACAGCGCGCC *CGG* .TT.....T..... ... Enzymes: *MspI, LpnPI, HinP1I, BspFNI, BstC8I, BssHII, NciI, StyD4I, MwoI* Cloning / PCR primers	90	94	43	54	-31	71	82	0 - 0 - 0 - 8 - 73 0 - 0 - 0 - 1 - 0 81 off-targets	3:exon:PNMAL2/AC011484.1 4:intergenic:AC105761.1/RNU6-685P-Y_RNA 4:intergenic:EIF4A3-CARD14 show all...
36 / rev	CCGAGTCCTTAGGCCGCCAG *GGG* ..G...C..... ... Enzymes: *LpnPI, HinP1I, AcyI, NlaIV, BseDI, BfoI, SspDI, BanI, BstNI, StyD4I, BslI* Cloning / PCR primers	87	95	56	95	44	73	84	0 - 0 - 0 - 5 - 49 0 - 0 - 0 - 0 - 0 54 off-targets	4:intergenic:ZCCHC7-RP11-397D12.4 4:intergenic:RP11-438O11.1-RPS7P5 4:intergenic:C10orf55-VCL show all...
76 / fw	CGGCGCGCCGGAAGTGGCCA *GGG* G.....C ... ⚠High GC content Enzymes: *LpnPI, BshFI, AcoI, BseDI, BalI, BstNI, StyD4I, BslI* Cloning / PCR primers	87	92	52	47	-18	56	72	0 - 0 - 1 - 5 - 51 0 - 0 - 0 - 0 - 2 57 off-targets	4:intron:LINC00925 4:intergenic:MNX1-AS1-RP4-639J15.1 4:intergenic:RP5-983L19.2-RN7SKP252 show all...
13 / rev	GCGCCGGCGCGCGCCCAGAT *TGG* .....T..... ... ⚠High GC content Enzymes: *LpnPI, BslFI, BslI* Cloning / PCR primers	86	97	20	78	-98	72	84	0 - 0 - 0 - 14 - 163 0 - 0 - 0 - 0 - 0 177 off-targets	4:exon:CTD-2623N2.11 4:intergenic:XRCC6-NHP2L1 4:intergenic:KIAA1644-RP1-32I10.10 show all...
144 / rev	GCCGTTGTCGACGACGAGCG *CGG* T..... ... Enzymes: *HinP1I, BspFNI, Fsp4HI, TauI, PcsI* Cloning / PCR primers	86	95	62	59	42	38	84	0 - 1 - 1 - 9 - 10 0 - 0 - 0 - 0 - 1 21 off-targets	3:exon:ACTBP7 2:exon:ACTBP2 3:exon:POTEG show all...
4 / rev	CGCGCCCAGATTGGGGACAA *AGG* .... Enzymes: *BslFI* Cloning / PCR primers	85	94	51	76	-22	71	79	0 - 0 - 1 - 4 - 60 0 - 0 - 0 - 1 - 0 65 off-targets	4:intron:ENTPD2 4:intron:FBXL18 4:intergenic:Y_RNA-EXOC6 show all...
56 / fw	CCCCTGGCGGCCTAAGGACT *CGG* .....C... G.. Enzymes: *HinP1I, BspFNI, BstC8I, BssHII, HinfI* Cloning / PCR primers	83	91	33	60	-27	64	74	0 - 0 - 0 - 9 - 98 0 - 0 - 0 - 0 - 4 107 off-targets	4:intergenic:ARTN-RP11-7O11.3 4:exon:SLC17A9 4:intergenic:HAUS1P2-PHKA2-AS1 show all...
35 / rev	CGAGTCCTTAGGCCGCCAGG *GGG* .G...C..... ... Enzymes: *LpnPI, HinP1I, AcyI, NlaIV, BseDI, BfoI, SspDI, BanI, BstNI, StyD4I, BslI* Cloning / PCR primers	82	92	65	64	105	71	85	0 - 0 - 3 - 7 - 64 0 - 0 - 0 - 0 - 1 74 off-targets	4:intergenic:RP11-268I9.3-RP11-268I9.4 4:intron:STK32B 3:intergenic:RP11-438O11.1-RPS7P5 show all...
129 / fw	GCTATTCTCGCAGCTCACCA *TGG* .....G.TT...T..... ... Enzymes: *NlaIII, ErhI, BseDI, Bsp19I, BstDSI, BtsCI* Cloning / PCR primers	79	93	55	34	106	61	83	0 - 0 - 1 - 16 - 62 0 - 0 - 0 - 1 - 3 79 off-targets	4:intron:SYT16 4:intergenic:RPS6KC1-RPL31P13 3:intergenic:CTD-2089N3.3/CTD-2089N3.2-RNU6-480P show all...
50 / fw	CGGCGCCCCCTGGCGGCCTA *AGG* T..... ..C ⚠High GC content Enzymes: *Bse21I, BshFI, HinfI, BstDEI* Cloning / PCR primers	77	87	29	68	-97	66	81	0 - 0 - 3 - 23 - 135 0 - 0 - 0 - 1 - 0 161 off-targets	4:intron:MAD1L1 4:intron:JADE2 2:exon:GDF15 show all...
106 / rev	TGGTGAGCTGCGAGAATAGC *CGG* .....T...TT.G..... ... Enzymes: *MspI, LpnPI, HinP1I, BspFNI, NciI, StyD4I, MwoI* Cloning / PCR primers	77	92	43	54	-33	72	76	0 - 0 - 1 - 9 - 109 0 - 0 - 0 - 0 - 1 119 off-targets	4:intergenic:MEPE-HSP90AB3P 4:exon:ASIC2 3:intergenic:RP11-415C15.2-IGBP1P5 show all...
11 / rev	GCCGGCGCGCGCCCAGATTG *GGG* ...T..... ... ⚠High GC content Enzymes: *BslFI, BslI* Cloning / PCR primers	76	89	44	53	81	71	81	0 - 1 - 0 - 7 - 74 0 - 0 - 0 - 0 - 0 82 off-targets	3:intergenic:RP5-1097P24.1-RP5-828K20.1 3:intergenic:AC004870.3-TNS3 4:intergenic:RN7SL701P-AJ239322.3 show all...
79 / fw	CGCGCCGGAAGTGGCCAGGG *CGG* .....C... ...	75	86	43	93	11	55	76	0 - 0 - 0 - 10 - 94 0 - 0 - 0 - 0 - 8	4:intron:PRDM16

Position/ Strand	Guide Sequence + *PAM* + Restriction Enzymes + Variants Only G- Only GG- Only A-	MIT Specificity Score	CFD Spec. score	Predicted Efficiency Show all scores Doench '16	Mor.-Mateo	Doench-RuleSet3	Out-of-Frame	Lindel	Off-targets for 0-1-2-3-4 mismatches + next to PAM	Genome Browser links to matches sorted by CFD off-target score exons only chr7 only
	⚠High GC content Enzymes: *LpnPI, BseDI, BstNI, StyD4I, BslI* Cloning / PCR primers								104 off-targets	3:intergenic:NOMO1-MIR3179-1 3:intergenic:RP11-517A5.4/NOMO3-MIR3179-2 show all...
88 / rev	GCCGGGCGCGCTGTGAGCCG *AGG* .....T..... .TT ⚠High GC content Enzymes: *NmeAIII, BseDI, MwoI* Cloning / PCR primers	74	86	57	60	52	57	79	0 - 0 - 2 - 22 - 177 0 - 0 - 1 - 1 - 1 201 off-targets	4:intergenic:RP11-616M22.2-CACNA1H 4:exon:MAP3K10 4:intron:CDC25C show all...
105 / rev	GGTGAGCTGCGAGAATAGCC *GGG* .....T...TT.G..... ... ⚠Inefficient Enzymes: *MspI, LpnPI, HinP1I, BspFNI, BstC8I, BssHII, NciI, StyD4I, MwoI* Cloning / PCR primers	73	88	59	46	5	73	70	0 - 0 - 4 - 29 - 120 0 - 0 - 0 - 0 - 0 153 off-targets	3:intron:NTNG1 2:intron:C1orf170 4:intron:ZFP82 show all...
73 / rev	AGCCGAGGTCGCCCCCGCCC *TGG* .....TT..... C.. ⚠High GC content Enzymes: *LpnPI, BshFI, SsiI, AcoI, BseDI, BalI, FauI, BstNI, StyD4I, BslI* Cloning / PCR primers	69	79	43	42	-18	56	66	0 - 0 - 1 - 15 - 183 0 - 0 - 0 - 1 - 3 199 off-targets	4:intron:SPACA3 4:exon:CCDC159 4:exon:HHIP-AS1 show all...
63 / rev	GCCCCCGCCCTGGCCACTTC *CGG* .....C..... ... ⚠Inefficient ⚠High GC content Enzymes: *MspI, LpnPI, HinP1I, BspFNI, BstC8I, BssHII* Cloning / PCR primers	63	81	30	36	-70	59	74	0 - 0 - 3 - 27 - 271 0 - 0 - 0 - 6 - 6 301 off-targets	4:intergenic:U3-HSPA8P8 4:intron:CTRB1 4:intergenic:RP11-98L5.4-RP11-98L5.5 show all...
81 / fw	CGCCGGAAGTGGCCAGGGCG *GGG* .....C..... ... ⚠High GC content Enzymes: *BseDI, BslI* Cloning / PCR primers	61	82	23	57	-72	57	70	0 - 0 - 2 - 18 - 152 0 - 0 - 0 - 1 - 6 172 off-targets	4:intergenic:CTD-2307P3.1-RP11-97N10.1 3:intergenic:SNAI1-TRERNA1 4:intron:WWP2 show all...
29 / rev	CTTAGGCCGCCAGGGGGCGC *CGG* ..... .T. ⚠High GC content Enzymes: *MspI, LpnPI, HinP1I, BspFNI, AcyI, NlaIV, BstC8I, Cfr10I, BssHII, BfoI, NaeI, SspDI, BanI, SgrAI, MreI* Cloning / PCR primers	60	85	34	55	-9	63	74	0 - 0 - 2 - 22 - 128 0 - 0 - 1 - 3 - 7 152 off-targets	4:exon:ANXA11 3:intergenic:FGF18-AC011410.1 4:intron:PITPNM2 show all...
82 / fw	GCCGGAAGTGGCCAGGGCGG *GGG* .....C..... ... ⚠High GC content Enzymes: *BslI* Cloning / PCR primers	60	67	32	86	51	53	72	0 - 0 - 2 - 34 - 293 0 - 0 - 0 - 2 - 4 329 off-targets	3:intron:TMEM194A 4:intron:FAM227A 2:exon:TPD52L2 show all...
91 / fw	GGCCAGGGCGGGGGCGACCT *CGG* ....C.....TT. ... ⚠High GC content Enzymes: *MnlI, BseDI, MwoI* Cloning / PCR primers	59	85	42	71	-78	57	64	0 - 0 - 3 - 33 - 327 0 - 0 - 0 - 0 - 3 363 off-targets	4:intron:LDB3 4:intron:RP11-1055B8.7 4:exon:CYP2W1 show all...
80 / fw	GCGCCGGAAGTGGCCAGGGC *GGG* .....C..... ... ⚠High GC content Enzymes: *BseDI, BstNI, StyD4I, BslI* Cloning / PCR primers	52	79	28	67	-17	56	64	0 - 0 - 3 - 26 - 218 0 - 0 - 0 - 6 - 14 247 off-targets	4:intergenic:RP4-614C15.3-PIEZO1P1 3:intergenic:RP11-16E18.1-YTHDF3-AS1 4:intergenic:AL390778.1-C9orf62 show all...
40 / fw	GGGCGCGCGCCGGCGCCCCC *TGG* .....T..... ... ⚠High GC content Enzymes: *LpnPI, BshFI, BseDI, Fsp4HI, BstNI, StyD4I, TauI, BslI* Cloning / PCR primers	48	77	34	60	-146	75	80	0 - 0 - 8 - 60 - 361 0 - 0 - 1 - 7 - 2 429 off-targets	3:intergenic:AL592294.1-ZSWIM5 3:intron:WBSCR17 3:intergenic:SRSF9-DYNLL1 show all...
169 / fw	CTCGTCGTCGACAACGGCTC *CGG* .....T.. ... Enzymes: *MspI, LpnPI, NlaIII, XceI, NlaIV, BceAI, LmnI* Cloning / PCR primers	43	89	52	59	43	38	52	0 - 1 - 13 - 4 - 13 0 - 1 - 0 - 0 - 2 31 off-targets	2:exon:POTEG 2:exon:POTEM 2:exon:POTEF show all...
183 / rev	GGCATCGTCGCCCGCGAAGC *CGG* .... Enzymes: *MspI, LpnPI, BshFI, BstC8I, Cfr10I, NaeI, MwoI* Cloning / PCR primers	41	89	39	43	-33	45	70	0 - 6 - 5 - 6 - 29 0 - 0 - 0 - 0 - 2 46 off-targets	1:exon:POTEF 1:exon:POTEI/AC013269.1 1:exon:POTEJ/AC140481.1 show all...
43 / fw	CGCGCGCCGGCGCCCCCTGG *CGG* .....T..... ... ⚠High GC content Enzymes: *LpnPI, BshFI, BseDI, Fsp4HI, BstNI, StyD4I, TauI, BslI* Cloning / PCR primers	40	73	47	77	0	69	85	0 - 0 - 12 - 30 - 263 0 - 0 - 5 - 6 - 37 305 off-targets	2:intron:SALL4 4:intron:BOP1 4:exon:LHX5 show all...

Position/ Strand	Guide Sequence + *PAM* + Restriction Enzymes + Variants Only G- Only GG- Only A-	MIT Specificity Score	CFD Spec. score	Predicted Efficiency (Show all scores): Doench '16	Mor.-Mateo	Doench-RuleSet3	Out-of-Frame	Lindel	Off-targets for 0-1-2-3-4 mismatches + next to PAM	Genome Browser links to matches sorted by CFD off-target score exons only chr7 only
126 / rev	CGCGGCGATATCATCATCCA *TGG* Enzymes: *NlaIII, ErhI, BseDI, Bsp19I, BstDSI, AsuHPI* Cloning / PCR primers	32	88	55	39	34	61	88	0 - 2 - 6 - 8 - 20 0 - 2 - 4 - 8 - 4 36 off-targets	1:exon:ACTBP2 2:exon:RP11-12M9.4 2:exon:POTEI/AC013269.1 show all...
205 / rev	TGGAGGGGAAGACGGCCCGG *GGG* Enzymes: *MspI, LpnPI, BmeT110I, Cfr9I, BseDI, PspPI, LweI, NciI, StyD4I, MwoI, BglI* Cloning / PCR primers	30	73	52	97	71	62	78	1 - 4 - 12 - 36 - 199 1 - 0 - 0 - 0 - 0 252 off-targets	0:intron:FNIP1 1:exon:ACTG2 4:intron:COL20A1 show all...
204 / rev	GGAGGGGAAGACGGCCCGGG *GGG* ⚠ High GC content Enzymes: *MspI, LpnPI, BmeT110I, Cfr9I, BseDI, LweI, NciI, StyD4I, MwoI, BglI* Cloning / PCR primers	29	69	52	101	1	57	61	0 - 5 - 17 - 37 - 356 0 - 0 - 0 - 1 - 1 415 off-targets	1:exon:ACTG2 2:exon:POTEG 2:exon:POTEM show all...
206 / rev	ATGGAGGGGAAGACGGCCCG *GGG* Enzymes: *MspI, LpnPI, BshFI, BmeT110I, Cfr9I, BseDI, PspPI, NciI, StyD4I, MwoI, BglI* Cloning / PCR primers	29	78	40	87	-7	59	79	1 - 4 - 14 - 22 - 125 1 - 0 - 0 - 0 - 0 166 off-targets	0:intron:FNIP1 1:exon:ACTG2 2:exon:POTEG show all...
192 / fw	CATGTGCAAGGCCGGCTTCG *CGG* ⚠ Inefficient Enzymes: *BspFNI, BstC8I, MwoI* Cloning / PCR primers	25	88	60	46	-1	50	73	1 - 10 - 6 - 5 - 71 1 - 0 - 0 - 0 - 0 93 off-targets	0:exon:RP11-12M9.4 2:exon:ACTBP2 4:intergenic:RNA5SP99-AC013262.1 show all...
208 / rev	CGATGGAGGGGAAGACGGCC *CGG* ⚠ Inefficient Enzymes: *MspI, LpnPI, BshFI, BmeT110I, Cfr9I, BseDI, BceAI, PspPI, NciI, StyD4I, MwoI, BglI* Cloning / PCR primers	24	74	37	47	-61	56	79	1 - 4 - 12 - 21 - 165 1 - 2 - 1 - 0 - 3 203 off-targets	1:exon:RP11-390F4.8 2:exon:POTEG 2:exon:POTEM show all...
193 / fw	ATGTGCAAGGCCGGCTTCGC *GGG* Enzymes: *BspFNI, BstC8I, Hpy99I, MwoI* Cloning / PCR primers	23	83	41	54	-82	46	64	1 - 10 - 8 - 14 - 83 1 - 0 - 0 - 0 - 0 116 off-targets	0:exon:RP11-12M9.4 2:exon:RP11-809F4.2 1:exon:CH17-12M21.1 show all...
231 / fw	GGCCGTCTTCCCCTCCATCG *TGG* Enzymes: *HinP1I, BccI, BslI* Cloning / PCR primers	23	82	53	41	65	70	76	1 - 3 - 14 - 17 - 100 1 - 1 - 0 - 1 - 3 135 off-targets	0:exon:AC139143.2 1:exon:CTC-543D15.1 3:intergenic:CTC-565M22.1-ZNF536 show all...
168 / rev	GAAGCCGGCCTTGCACATGC *CGG* Enzymes: *MspI, LpnPI, NlaIII, XceI, NlaIV* Cloning / PCR primers	21	81	53	49	13	39	65	2 - 11 - 7 - 22 - 97 2 - 11 - 2 - 1 - 1 139 off-targets	1:exon:ACTBP11 1:exon:ACTBP2 3:exon:AMY2B/ACTG1P4 show all...
207 / rev	GATGGAGGGGAAGACGGCCC *GGG* Enzymes: *MspI, LpnPI, BshFI, BmeT110I, Cfr9I, BseDI, BceAI, PspPI, NciI, StyD4I, MwoI, BglI* Cloning / PCR primers	21	77	33	77	-6	57	74	1 - 4 - 16 - 32 - 269 1 - 2 - 0 - 1 - 0 322 off-targets	0:exon:AC139143.2 1:intron:FNIP1 2:exon:POTEG show all...
233 / fw	CCGTCTTCCCCTCCATCGTG *GGG* Enzymes: *BslI* Cloning / PCR primers	20	82	64	60	52	69	77	1 - 4 - 12 - 14 - 102 1 - 1 - 0 - 0 - 1 133 off-targets	0:exon:AC139143.2 1:exon:CTC-543D15.1 1:exon:RP11-809F4.2 show all...
209 / fw	TCGCGGGCGACGATGCCCCC *CGG* ⚠ High GC content Enzymes: *MspI, LpnPI, BshFI, BmeT110I, Cfr9I, BseDI, PspPI, NciI, StyD4I, MwoI, BglI* Cloning / PCR primers	18	88	36	68	0	59	77	0 - 5 - 4 - 12 - 26 0 - 5 - 0 - 4 - 1 47 off-targets	1:exon:POTEF 1:exon:POTEI/AC013269.1 1:exon:POTEJ/AC140481.1 show all...
232 / fw	GCCGTCTTCCCCTCCATCGT *GGG* Enzymes: *BccI* Cloning / PCR primers	18	79	51	35	45	69	70	1 - 4 - 17 - 18 - 86 1 - 2 - 0 - 0 - 0 126 off-targets	0:exon:AC139143.2 1:exon:CTC-543D15.1 1:exon:ACTBP2 show all...
213 / rev	CCCCACGATGGAGGGGAAGA *CGG* Enzymes: *MspI, LpnPI, BshFI, BbsI, BceAI, PspPI, MboII, NciI, StyD4I* Cloning / PCR primers	17	62	50	46	-40	60	61	2 - 10 - 14 - 55 - 248 2 - 1 - 4 - 22 - 13 329 off-targets	0:exon:AC139143.2 0:exon:CTC-543D15.1 1:exon:ACTBP8 show all...
210 / fw	CGCGGGCGACGATGCCCCCC *GGG* ⚠ High GC content Enzymes: *MspI, LpnPI, BshFI, BmeT110I, Cfr9I, BseDI, PspPI, NciI, StyD4I, MwoI,*	16	88	40	78	-9	57	60	0 - 5 - 1 - 8 - 78 0 - 5 - 0 - 4 - 2 92 off-targets	1:exon:POTEF 1:exon:POTEI/AC013269.1 1:exon:POTEJ/AC140481.1 show all...

Position/ Strand	Guide Sequence + *PAM* + Restriction Enzymes + Variants Only G- Only GG- Only A-	MIT Specificity Score	CFD Spec. score	Predicted Efficiency Show all scores Doench '16	Mor-Mateo	Doench-RuleSet3	Out-of-Frame	Outcome Lindel	Off-targets for 0-1-2-3-4 mismatches + next to PAM	Genome Browser links to matches sorted by CFD off-target score exons only chr7 only
	BglI **Cloning / PCR primers**									
184 / fw	GGCTCCGGCATGTGCAAGGC *CGG* ..I..... Enzymes: *MspI, LpnPI, BshFI, BstC8I, Cfr10I, NaeI, MwoI* **Cloning / PCR primers**	11	74	47	42	-34	44	73	1 - 14 - 11 - 22 - 139 1 - 12 - 2 - 3 - 2 187 off-targets	1:exon:POTEG 1:exon:POTEM 1:exon:POTEF show all...
180 / fw	CAACGGCTCCGGCATGTGCA *AGG* .....I..... Enzymes: *MspI, LpnPI, BshFI, HpyCH4V, BstC8I, Cfr10I, NaeI, MwoI* **Cloning / PCR primers**	7	81	56	35	40	45	81	1 - 14 - 11 - 14 - 51 1 - 3 - 0 - 0 - 0 91 off-targets	0:exon:ACTBP2 1:exon:POTEG 1:exon:POTEM show all...

Download as Excel tables: Guides / Guides, all scores / Off-targets / Saturating mutagenesis assistant
Tab-sep format: Guides / Off-targets

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